Seminar 05 cheat sheet

Second-Order and Quasi-Newton Methods

Optimization Methods
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Newton step

- $m(p) = f + g^\top p + \frac{1}{2} p^\top H p$; $Hp_N = -g$, $x^+ = x + p_N$.
- Solve by Cholesky or LU, never with H^{-1}.
- Quadratic, $Q \succ 0$: one step, $x^+ = Q^{-1} b$.
- $\nabla^2 f(x^*) \succ 0$, M-Lipschitz H: $\|e^+\| \le \frac{M}{2m} \|e\|^2$.
- Singular $H(x^*)$: only linear, e.g. x^4 gives $x^+ = \frac{2}{3} x$.

Failure modes

- $g^\top p_N = -g^\top H^{-1} g$: may be > 0 if H is indefinite.
- Newton solves $\nabla f = 0$: saddles and maxima attract.
- Good model at x, step outside its region: f may jump up.

Safeguards

- Armijo: $f(x + \alpha p) \le f + c\alpha g^\top p$, $\alpha = 1, \beta, \beta^2, \ldots$
- $c < \frac{1}{2}$, $H(x^*) \succ 0$: eventually $\alpha = 1$, quadratic rate kept.
- $(H + \lambda I)p = -g$, $\lambda > -\lambda_{\min}(H)$; raise λ until Cholesky works.
- $\lambda \to 0$: Newton; $\lambda \to \infty$: $p \approx -g/\lambda$.

Least squares

- $f = \frac{1}{2}\|r\|^2$: $\nabla f = J^\top r$, $\nabla^2 f = J^\top J + \sum r_i \nabla^2 r_i$.
- Gauss–Newton $J^\top J p = -J^\top r$; LM $(J^\top J + \lambda I)p = -J^\top r$.

Cost

- Dense: $O(n^2)$ memory, $\approx n^3/3$ for Cholesky.
- Logistic: $Hv = N^{-1} X^\top (d \odot Xv) + \gamma v$, $d = \sigma(1 - \sigma)$.
- Newton-CG: k products cost $\approx 2kNn$ vs Nn^2 to form H.

BFGS

- $s = x^+ - x$, $y = g^+ - g$; $B^+ s = y$, $G^+ y = s$.
- $G^+ = (I - \rho s y^\top) G (I - \rho y s^\top) + \rho s s^\top$, $\rho = 1/y^\top s$.
- $B^+ = B - \frac{B s s^\top B}{s^\top B s} + \frac{y y^\top}{y^\top s}$; $p = -Gg$.
- $G \succ 0$, $y^\top s > 0 \Rightarrow G^+ \succ 0$. Superlinear near x^*.
- Exact line search on an n-dim quadratic: at most n steps.

Curvature condition

- $y^\top s$ = average curvature along s; $\ge \mu \|s\|^2$ if strongly convex.
- Wolfe: $\nabla f(x + \alpha p)^\top p \ge c_2 g^\top p \Rightarrow y^\top s > 0$.
- Skip if $y^\top s \le \varepsilon \|y\| \|s\|$, or damp.

L-BFGS

- Keep m pairs; $q = g$; newest to oldest: $a_i = \rho_i s_i^\top q$, $q \mathrel{-}= a_i y_i$.
- $r = \gamma q$, $\gamma = \frac{s^\top y}{y^\top y}$ (last pair).
- Oldest to newest: $b_i = \rho_i y_i^\top r$, $r \mathrel{+}= s_i (a_i - b_i)$; $p = -r$.
- $\approx 4mn$ flops, $2mn$ numbers. Bounds: L-BFGS-B, not clipping.

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